

An Exact Neumann-Mode Turing Atlas for the Schnakenberg Reaction–Diffusion System

Route-A source-local certificate HCS-C350

3 September 2026

Abstract

For the Schnakenberg system on a finite interval we derive the unique positive constant equilibrium, its kinetic stability chamber, and the exact two-by-two dispersion matrix for every Neumann mode.

1 Frozen model and kinetic chamber

Let $a, b, d_u, d_v, L > 0$ and consider, on $0 < x < L$,

$$\begin{aligned} u_t &= d_u u_{xx} + a - u + u^2 v, \\ v_t &= d_v v_{xx} + b - u^2 v, \end{aligned} \quad u_x = v_x = 0 \quad \text{at } x = 0, L. \quad (1)$$

Put $s = a + b$. A positive spatially homogeneous equilibrium must satisfy $u^2 v = b$ and $a - u + b = 0$. Thus the unique positive spatially homogeneous equilibrium is

$$(u_*, v_*) = (s, b/s^2). \quad (2)$$

The reaction Jacobian there is

$$J = \begin{pmatrix} (b-a)/s & s^2 \\ -2b/s & -s^2 \end{pmatrix}, \quad \tau := \text{tr } J = \frac{b-a}{s} - s^2, \quad \det J = s^2. \quad (3)$$

Thus the kinetic equilibrium is asymptotically stable exactly when $\tau < 0$. At $\tau = 0$ its eigenvalues are $\pm is$; for $\tau > 0$ it has positive spectral abscissa.

Expand perturbations in the normalized Neumann cosine basis. With

$$\mu_n = (n\pi/L)^2, \quad n = 0, 1, 2, \dots, \quad (4)$$

the n th coefficient evolves under

$$M(\mu) = J - \begin{pmatrix} d_u \mu & 0 \\ 0 & d_v \mu \end{pmatrix}. \quad (5)$$

Direct expansion gives

$$T(\mu) = \text{tr } M(\mu) = \tau - (d_u + d_v)\mu, \quad (6)$$

$$D(\mu) = \det M(\mu) = d_u d_v \mu^2 - B\mu + s^2, \quad B := d_v \frac{b-a}{s} - d_u s^2. \quad (7)$$

Consequently the complete modal pair is

$$\lambda_{n,\pm} = \frac{T(\mu_n) \pm \sqrt{T(\mu_n)^2 - 4D(\mu_n)}}{2}. \quad (8)$$

The phrase *kinetic and modal dispersion closure* marks the baseline manuscript: no continuum-wavenumber test has yet been promoted to a finite interval claim.